CPU-ACA Form No. 25
Rev. 03, July 2022

Central Philippine University
COLLEGE OF COMPUTER STUDIES

BACHELOR OF SCIENCE IN COMPUTER SCIENCE
Based on CHED Memo Order No. 25 s. 2015
Effective: Academic Year 2024-2025

FIRST YEAR

First Semester

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 1001	Introduction to Computing	2	3	3	
	CCS 1400	Fundamentals of Programming	2	3	3	
	GEMath 1	Mathematics in the Modern World	3	0	3	
	GESocSci 5	Ethics	3	0	3	
	CETech 1	Living in the IT Era	3	0	3	*
	GESocSci 2	Readings in Philippine History	3	0	3	
	SEAL 1	Student Enhancement Activities for Life I	1	0	1	
	RE 1	Christianity in a Changing Society	3	0	3	
	PATHFit1 W	Physical Activity Toward Health and Fitness (PATHFit)- Movement Competency Training	(2)	0	(2)	
	PATHFit1 M					
	NSTP 1 – CWTS LTS ROTC	Civic Welfare Training Service Literacy Training Service Reserve Officers' Training Corps	(3)	0	(3)	
		SUB-TOTAL ACADEMIC UNITS	**25**	**6**	**27**	

Second Semester

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 1500	Intermediate Programming	2	3	3	CCS 1001, CCS 1400
	CCS 1301	Data Structures and Algorithms	2	3	3	CCS 1400
	GESocSci 4	The Contemporary World	3	0	3	
	SEAL 2	Student Enhancement Activities for Life II	1	0	1	
	GEEng 1	Purposive Communication	3	0	3	
	GESocSci 1	Understanding the Self	3	0	3	
	CESocSci 4	The Entrepreneurial Mind	3	0	3	
	RE 2	Christian Ethics in a Changing World	3	0	3	RE 1
	PATHFit2 W	Physical Activity Toward Health and Fitness (PATHFit)- Exercise Based Fitness Activities	(2)	0	(2)	PATHFit1 W
	PATHFit2 M					PATHFit1 M
	NSTP 2 – CWTS LTS ROTC	Civic Welfare Training Service Literacy Training Service Reserve Officers' Training Corps	(3)	0	(3)	NSTP 1
		SUB-TOTAL ACADEMIC UNITS	**25**	**6**	**27**	

SECOND YEAR

First Semester

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 2100	Fundamentals of Database Design	2	3	3	CCS 1301, CCS 1500
	CS 2111	Structure of Programming Languages	3	0	3	CCS 1301, CCS 1500
	CCS 2110	Application Development and Emerging Technologies	3	0	3	CCS 1500, *
	CCS 2200	Basic Electrical and Electronic Concepts	2	3	3	Passed all 1st Yr. CS Subjects
	Math 2110	Calculus	3	0	3	GEMath 1
	CS 2120	Discrete Structures I	3	0	3	GEMath 1
	CCS 2801	Mobile Application Development I	0	3	1	CCS 1500, *
	GEHum 1	Art Appreciation	3	0	3	
	PATHFit3 W	Physical Activity Toward Health and Fitness (PATHFit)-Dance and Swimming	(2)	0	(2)	PATHFit2 W
	PATHFit3 M					PATHFit2 M
		SUB-TOTAL ACADEMIC UNITS	**21**	**9**	**24**	

Second Semester

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 2300	Logic Design	2	3	3	CCS 2200
	CCS 2401	Network Engineering I: Introduction to Networks	2	3	3	CCS 2200
	CCS 2501	System Analysis and Design	3	0	3	CCS 2100, *
	CCS 2601	Programming with Databases	3	0	3	CCS 2100, *
	IT 3110	Computer Hardware Repair and Maintenance	2	3	3	CCS 2200, (CCS 2401)
	CS 2210	Discrete Structures II	3	0	3	CS 2120
	CCS 3801	Mobile Application Development II	0	3	1	CCS 2801, *
	CEArts 3	Reading Visual Art	3	0	3	
	PATHFit4 W	Physical Activity Toward Health and Fitness (PATHFit)-Volleyball and Basketball	(2)	0	(2)	PATHFit3 W
	PATHFit4 M					PATHFit3 M
		SUB-TOTAL ACADEMIC UNITS	**20**	**12**	**24**	

THIRD YEAR

First Semester

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 3002	Computer Organization & Assembly Language	0	3	1	CCS 2300, *
	CCS 3010	Fundamentals of Human Computer Interaction	3	0	3	CCS 2501, *
	CCS 3020	Information Assurance and Security I	3	0	3	CCS 2401
	CCS 3031	Web Systems and Technologies	0	3	1	CCS 2601, CCS 2110
	CCS 3100	Methods of Research in IT	3	0	3	Passed all 2nd Yr. 2nd Sem. CS Subjects
	CSPE 1	Professional Elective	3	0	3	Passed all 2nd Yr. CS Subjects, *
	CS 3120	Algorithms and Complexities	3	0	3	CS 2111, *
	GESocSci 3	Life and Works of Rizal	3	0	3	
	GESocSci 6	Science, Technology and Society	3	0	3	
		SUB-TOTAL ACADEMIC UNITS	**21**	**6**	**23**	

Second Semester

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 3300	Software Engineering	3	0	3	CCS 3100, *
	CCS 3501	Operating Systems	3	0	3	CCS 3001, *
	CCS 3600	CCS Thesis I	2	3	3	Passed all 3rd Yr. 1st Sem. CS Subjects
	CS 3110	Automata Theory & Computability	3	0	3	CS 2111
	CS 3210	Computer Graphics and Visual Computing	2	3	3	CCS 3010
	CSPE 2	Professional Elective	3	0	3	CSPE 1, *
		SUB-TOTAL ACADEMIC UNITS	**16**	**6**	**18**	

SUMMER YEAR

Grade	Subject Code	Subject Title	Lec.hrs/ week	Lab.hrs/ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 4001	Seminars	3	0	3	Passed all 3rd Yr. CS Subjects
		SUB-TOTAL ACADEMIC UNITS	**3**	**0**	**3**	

FOURTH YEAR

First Semester

Grade	Subject Code	Subject Title	Lec.hrs. /week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 4100	CCS Thesis II	2	3	3	CCS 3600
	CS 4110	Artificial Intelligence	2	3	3	CS 3110
	CCS 4300	Social Issues and Professional Practices	3	0	3	
	CSPE 3	Professional Elective	3	0	3	CSPE 2, *
		SUB-TOTAL ACADEMIC UNITS	**10**	**6**	**12**	

Second Semester

Grade	Subject Code	Subject Title	Lec.hrs. / week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CCS 4201	On-the-Job Training (600 hours)	6	0	6	Passed ALL Subjects
		SUB-TOTAL ACADEMIC UNITS	**6**	**0**	**6**	
		GRAND TOTAL ACADEMIC UNITS			**164**	

AREA OF SPECIALIZATION IN COMPUTER SCIENCE
PROFESSIONAL ELECTIVES (9 Units Required)

Software Development

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CSPE 4100	Software Development 1	3	0	3	Passed all 2nd Yr. 2nd Sem. CS Subjects, *
	CSPE 4200	Software Development 2	3	0	3	CCSPE 2101, *
	CSPE 4300	Software Development 3	3	0	3	CCSPE 2300, *
		TOTAL UNITS	**9**	**0**	**9**	

Data Science

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CSPE 3101	Data Science 1: Introduction to Data Science and Statistics	3	0	3	Passed all 2nd Yr. 2nd Sem. CS Subjects, *
	CSPE 3300	Data Science 3: Data and Network Security	3	0	3	CSPE 3101, *
	CSPE 3400	Data Science 4: Data Mining	3	0	3	CSPE 3300, *
		TOTAL UNITS	**9**	**0**	**9**	

Cybersecurity

Grade	Subject Code	Subject Title	Lec.hrs./ week	Lab.hrs./ week	Credit Units	Pre-requisites & (Co-requisites)
	CSPE 5100	Cybersecurity 1: Ethical Hacking	3	0	3	Passed all 2nd Yr. 2nd Sem. CS Subjects, *
	CSPE 5200	Cybersecurity 2: Ethics and Cyber Warfare	3	0	3	CSPE 5100 *
	CSPE 5300	Cybersecurity 3: Digital Forensics and Cybercrime	3	0	3	CSPE 5200, *
		TOTAL UNITS	**9**	**0**	**9**	

Legend: * - subject assigned in a laboratory with laboratory fees

Prepared by:

_____
MARJEE ROSE P. PARREÑO, MSCS
Program Coordinator, BSCS AND BSDMIA

Recommending Approval:

_____
MA. CHRISTINA A. QUIJANO, MSCS
Dean

_____
CHARISSA JUNE S. DEOCAMPO
Registrar

_____
IRVING DOMINGO L. RIO, D.Mgt
Vice President for Academic Affairs
Chairperson, University Curriculum Committee

APPROVED:

_____
ERNEST HOWARD B. DAGOHOY, M.Div
President

Date: _____